

Minimal Expansions Preserving Simple Random Walks

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Abstract

Let G be a finite connected simple graph with at least two vertices. A faithful expansion of G is a connected simple graph H equipped with a surjection onto G that sends simple random walk on H to simple random walk on G for every initial state. We prove that the minimum order of a nontrivial faithful expansion is $|V(G)| + \tau(G)$, where $\tau(G)$ is the vertex-cover number. Every such expansion has at least $2|E(G)|$ edges, and the two bounds are attained simultaneously. We characterize all order-minimizing expansions and show that their number, up to isomorphism over a fixed labeled base, is $\sum_S 2^{\beta(G[S])}$, where S ranges over the minimum vertex covers and β denotes cycle rank. Consequences include an exact formula for the cycle rank of an optimal expansion, preservation under time-dependent replacement, and an exponential edge bound for nested strict expansions. The arguments use classical degree conservation for harmonic morphisms. The role of unit edge weights is demonstrated by a contrasting weighted construction, and exhaustive finite checks accompany the proofs.

Keywords. Simple random walk; harmonic morphism; vertex cover; graph expansion; lumpability.

1 Introduction

Suppose that the vertices of a graph represent observable states, and that each state may be realized by several microscopic vertices. A natural requirement on such a realization is that a simple random walker, observed only through the original state labels, should retain its entire trajectory law. This requirement permits substantial changes in the underlying graph: for example, a four-cycle projects faithfully onto a three-vertex path by identifying opposite vertices. The resulting extremal question is how few vertices and edges a nontrivial realization can have.

For a graph G , let $\tau(G)$ be its vertex-cover number and $\alpha(G)$ its independence number. We show that every nontrivial faithful expansion H satisfies

$$|V(H)| \geq |V(G)| + \tau(G), \quad |E(H)| \geq 2|E(G)|,$$

with simultaneous equality attainable for every connected simple base. An optimal construction takes two copies of G and identifies the copies of a maximum independent set. More generally, all order-minimizing expansions arise from minimum vertex covers, with binary matching choices on the edges induced by the cover. These choices yield an exact enumeration over a fixed labeled base.

The probabilistic condition is a specialization of strong lumpability, which requires a projected Markov chain to be Markov with the same transition kernel for every initial distribution; see Geiger and Temmel [2]. For simple random walks on unweighted graphs, it is also the local-balance condition for a nondegenerate harmonic morphism without contracted edges. The corresponding

degree theory and its composition properties are classical; see Baker and Norine [1]. The present manuscript gives a self-contained extremal treatment of this setting. Historical priority of the minimum-order formulation and its enumeration has not been established, and no priority claim is made.

Section 2 fixes the model. Section 3 proves the extremal bounds, and Section 4 classifies their order-minimizing cases. Section 5 develops topological and temporal consequences. Section 6 treats the weighted variant and the scope of the observation model. Computational checks appear in Section 7.

2 Preliminaries and faithful expansions

All graphs in the main results are finite, connected, undirected, simple, and have at least two vertices. Edges have unit weight. Time is discrete: a simple random walk chooses one neighbor uniformly at each step. Write $d_G(v) = |N_G(v)|$ and

$$P_G(u, v) = \begin{cases} 1/d_G(u), & uv \in E(G), \\ 0, & \text{otherwise.} \end{cases}$$

Definition 2.1. A *faithful expansion* of G is a graph H together with a surjection $\phi : V(H) \rightarrow V(G)$ such that, for every starting vertex $x \in V(H)$, the projection of a simple random walk on H started at x has the same entire trajectory law as a simple random walk on G started at $\phi(x)$. The expansion is *nontrivial* if some fiber $\phi^{-1}(v)$ contains at least two vertices.

The term expansion refers to this definition; no induced-subgraph relation between G and H is required. The surjection identifies observable labels. Each expansion is a single replacement operation; an implementation by individual edge edits need not preserve the walk at intermediate stages.

The definition is equivalent to the following one-step condition:

$$\frac{|N_H(x) \cap \phi^{-1}(v)|}{d_H(x)} = P_G(\phi(x), v) \quad (x \in V(H), v \in V(G)). \quad (2.1)$$

Necessity follows by observing one step from each x . For sufficiency, condition on the entire observed past. Whatever the conditional distribution of the current hidden vertex, every vertex in its fiber has the same next-label distribution. Consequently, for any label sequence starting at $v_0 = \phi(x)$,

$$\mathbb{P}_x\{\phi(X_0) = v_0, \dots, \phi(X_t) = v_t\} = \prod_{i=0}^{t-1} P_G(v_i, v_{i+1}).$$

This proves preservation at every horizon and for every initial distribution, not only at stationarity. In matrix form the condition is $P_H C = C P_G$, where $C_{xv} = \mathbf{1}\{\phi(x) = v\}$.

3 Minimum order and size

A vertex cover meets every edge. Let $\tau(G)$ be its minimum size, and let $\alpha(G) = |V(G)| - \tau(G)$ be the maximum independent-set size.

Theorem 3.1 (Minimum expansion cost). *For every graph G as above,*

$$\min_{(H, \phi) \text{ nontrivial faithful}} |V(H)| = |V(G)| + \tau(G) = 2|V(G)| - \alpha(G).$$

Every nontrivial faithful expansion also satisfies

$$|E(H)| \geq 2|E(G)|.$$

There is one expansion attaining both bounds simultaneously.

The proof uses an integer conservation law. It is a special case of the degree theory of harmonic morphisms; see Baker–Norine, Lemmas 2.4 and 2.6 [1]. We give the argument directly for the present model.

Lemma 3.2 (Conservation of multiplicity). *For a faithful expansion there is a positive integer D such that every edge of G has exactly D edges of H above it. For each $x \in V(H)$ there is a positive integer k_x such that*

$$|N_H(x) \cap \phi^{-1}(v)| = k_x \quad (v \in N_G(\phi(x))), \quad (3.1)$$

$$d_H(x) = k_x d_G(\phi(x)), \quad (3.2)$$

$$\sum_{x \in \phi^{-1}(u)} k_x = D \quad (u \in V(G)). \quad (3.3)$$

In particular $|E(H)| = D|E(G)|$, and $D = 1$ precisely when ϕ is a graph isomorphism.

Proof. Equation (2.1) makes all neighbor counts toward the different neighbors of $\phi(x)$ equal to $d_H(x)/d_G(\phi(x))$. This is a positive integer k_x . The same equation forbids edges within a fiber or between fibers whose labels are nonadjacent in G .

Set $M(u) = \sum_{x \in \phi^{-1}(u)} k_x$. For an edge $uv \in E(G)$, count the edges between its two fibers from either side. Equation (3.1) gives

$$|E_H(\phi^{-1}(u), \phi^{-1}(v))| = M(u) = M(v).$$

Connectivity of G makes M constant; call its value D . Every edge of H lies above exactly one edge of G , proving the edge formula. If $D = 1$, every nonempty fiber consists of one vertex with $k_x = 1$, so ϕ is an isomorphism. The converse is immediate. $\square$

Proof of Theorem 3.1. Let $S = \{v : |\phi^{-1}(v)| \geq 2\}$ be the set of split labels. A nontrivial expansion has $D \geq 2$. If an edge uv had neither endpoint in S , both fibers would be singletons. Simplicity of H would permit at most one edge between them, contradicting $D \geq 2$. Thus S is a vertex cover, and

$$|V(H)| - |V(G)| = \sum_{v \in V(G)} (|\phi^{-1}(v)| - 1) \geq |S| \geq \tau(G).$$

The edge lower bound follows from Lemma 3.2.

For attainment, choose a minimum vertex cover S and its independent complement I . Take two copies of G and identify the two copies of each vertex in I . Keep both copies of each vertex in S , and let ϕ forget the copy index. No edges become duplicated, because I is independent. The resulting graph has $|V(G)| + |S|$ vertices and $2|E(G)|$ edges. It is connected: the two connected copies intersect at I , which is nonempty.

A vertex over S has exactly one neighbor of each required label, while a vertex over I has exactly two. Equation (2.1) holds in both cases, completing the construction and the proof. $\square$

4 Classification and enumeration

For any graph F , including a disconnected graph, put $\beta(F) = |E(F)| - |V(F)| + c(F)$, where $c(F)$ counts connected components, including isolated vertices. This is its cycle-space dimension.

Theorem 4.1 (Classification and count). *Fix a labeled base graph G . Every vertex-minimizing faithful expansion has the following form:*

1. Choose a minimum vertex cover S , and put $I = V(G) \setminus S$.
2. Give each label in I one vertex and each label in S two vertices.
3. Over an edge from I to S , connect the singleton to both copies.
4. Over an edge within S , choose either of the two perfect matchings between its two pairs of copies.

Every such choice is a connected faithful expansion. For a fixed S there are exactly $2^{\beta(G[S])}$ isomorphism classes over G , meaning that an isomorphism must preserve the original labels. Therefore the total is

$$\sum_{\substack{S \text{ vertex cover} \\ |S|=\tau(G)}} 2^{\beta(G[S])}.$$

This counts maps to the fixed labeled base; it does not count unlabeled source graphs or identify maps under automorphisms of G .

Proof. Equality in the vertex lower bound forces $|S| = \tau(G)$ and every split fiber to have exactly two vertices. Since I is nonempty, choose $u \in I$ and a neighbor $v \in S$. There can be at most two edges above uv , so $D \leq 2$. Nontriviality gives $D = 2$. Equation (3.3) now forces $k_x = 1$ on the two-vertex fibers and $k_x = 2$ on the singleton fibers. These local degrees give exactly the edge patterns stated in the theorem.

Conversely those patterns satisfy (2.1). To see connectivity, fix any $u \in I$. From any vertex of H , lift a base-graph path leading to u , choosing one available neighbor at every step. It ends at the unique vertex above u . Thus all vertices lie in one component.

For the count, assign copy indices 0, 1 over each vertex in S . Each edge of $G[S]$ receives one bit: 0 for parallel matching, 1 for crossed matching. Swapping the two copies of a vertex toggles exactly the bits of edges incident to it in $G[S]$. All isomorphisms over G consist of such swaps. There are $2^{|E(G[S])|}$ bit assignments. The group of $2^{|S|}$ vertex swaps has kernel consisting of swaps constant on each component, of size $2^{c(G[S])}$. Each orbit therefore has size $2^{|S|-c(G[S])}$, giving $2^{\beta(G[S])}$ orbits. Different choices of S cannot be isomorphic over G , since fiber sizes are preserved. $\square$

Example 4.2 (Two nonisomorphic optimal expansions of K_4). For $G = K_4$ and a fixed three-vertex minimum cover S , the induced graph $G[S]$ is a triangle, so there are two optimal expansions over that cover. One is two copies of K_4 glued at their remaining vertex. The other is a six-cycle with a new center adjacent to all six cycle vertices. Both have seven vertices and twelve edges, and both project to the same random walk on K_4 . They are nonisomorphic even without labels: they contain eight and six triangles respectively.

5 Topological and temporal consequences

5.1 Cycle rank and indistinguishability

The smallest example with a topological change is a path with three vertices. Its unique minimum vertex cover is the center. Splitting that center gives a four-cycle, as illustrated in Figure 1.

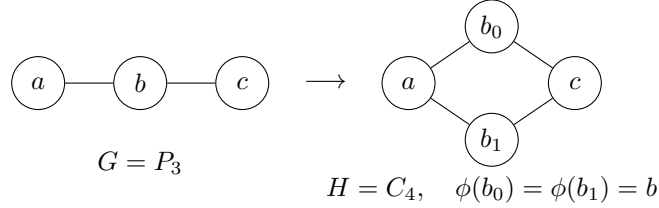


Figure 1: A faithful expansion of P_3 into C_4 . The two vertices b_0, b_1 share the observable label b ; the labels a, c each have one preimage. The projected walk has exactly the transition law on the path.

From a or c , the next observed label is always b . From either copy of b , the next label is a or c with probability $1/2$ each. An observer restricted to these labels cannot distinguish the trajectory laws at any horizon, even though one graph is a tree and the other has a cycle. An observer who can recognize individual microscopic vertices can distinguish them: a walk begun at b in P_3 returns to that very vertex in two steps with probability 1, whereas a walk begun at b_0 in C_4 returns to b_0 in two steps with probability $1/2$.

Corollary 5.1 (Hidden cycles in an optimal expansion). *For every vertex-minimizing expansion,*

$$\beta(H) = 2\beta(G) + \alpha(G) - 1.$$

In particular, if G is a tree, an optimal expansion has exactly $\alpha(G) - 1$ independent cycles.

Proof. Both graphs are connected, and optimality gives $|E(H)| = 2|E(G)|$ and $|V(H)| = 2|V(G)| - \alpha(G)$. Substitute these into $\beta(H) = |E(H)| - |V(H)| + 1$. $\square$

For a star with r leaves, splitting its center produces $K_{2,r}$. This uses only one additional vertex and creates $r - 1$ independent cycles.

5.2 Time-dependent replacement

Proposition 5.2 (Changing the graph between observations). *Let every H_t have a faithful projection ϕ_t onto the same base G . At each tick, move the current hidden vertex into any vertex of H_{t+1} with the same base label, then take one simple-random-walk step in H_{t+1} . The observed label sequence is exactly a simple random walk on G . The choice of H_{t+1} and the relocation may depend on the past, provided they occur before the fresh uniform neighbor draw.*

Proof. Relocation preserves the current label. Conditional on the entire past, the next label has distribution $P_G(u, \cdot)$ for every possible relocated vertex over u , by (2.1). Averaging over relocation and applying the chain rule proves the claim. $\square$

The relocation is between observed ticks; it is not counted as an additional walk step. Retaining the current base label is sufficient for preserving the observed transition law across each replacement.

5.3 Nested strict expansions

Corollary 5.3 (Cost of nested strict expansions). *Suppose each map in*

$$G_T \xrightarrow{\phi_T} G_{T-1} \longrightarrow \cdots \xrightarrow{\phi_1} G_0$$

is a nontrivial faithful expansion. Then the composite is faithful, and

$$|E(G_T)| = \left(\prod_{t=1}^T D_t \right) |E(G_0)| \geq 2^T |E(G_0)|.$$

Thus an edge budget $M \geq |E(G_0)|$ permits at most $\lfloor \log_2(M/|E(G_0)|) \rfloor$ successive strict expansions.

Proof. If $P_H C = C P_G$ and $P_G B = B P_F$, then $P_H C B = C B P_F$, proving composition. Lemma 3.2 supplies an integer factor $D_t \geq 2$ at each step; multiplying the edge counts gives the result. $\square$

The exponential cost concerns nested strict expansions. It does not apply to histories that alternate expansion and contraction, to relabelings, or to the arbitrary common-base replacements of Proposition 5.2.

6 Weighted expansions and scope

For a graph with positive edge weights $w_{xy} = w_{yx}$, the weighted walk chooses y from x with probability $w_{xy} / \sum_z w_{xz}$. Allowing such weights changes the minimum expansion cost.

Proposition 6.1 (Minimum order with adjustable weights). *For every unweighted base graph G in the main theorem, the smallest nontrivial faithful expansion allowing positive edge weights has $|V(G)| + 1$ vertices.*

Proof. Surjectivity and nontriviality require at least one additional vertex. Choose a vertex v and replace it by two copies. Replace each edge vu by two edges of weight $1/2$, one from each copy to u , and give every other edge weight 1. From u , the total weight toward label v remains 1. From either copy of v , all original neighboring labels remain equiprobable. All other transition probabilities are unchanged. The graph is connected and the expansion is faithful, proving attainment. $\square$

In particular, the vertex-cover bound depends on the unit-weight restriction. Simplicity is also used essentially when bounding the number of edges between singleton fibers in the proof of Theorem 3.1.

Likewise the observation restriction matters. The theorem preserves the discrete-time trajectory of the original labels, not the microscopic topology, individual-vertex return events, or a continuous-time walk with unit rate on every edge.

7 Computational verification

The accompanying program `verify_random_walk_refinements.py`, run with Python 3.13.7 and NetworkX 3.6.1, uses exact integer and rational arithmetic. Its saved output is `verification_results.json`.

The exhaustive source check examines every set partition of every connected Graph Atlas graph of orders 2–7, up to isomorphism of the source. This covers every surjection up to renaming

Table 1: Exact finite checks of the extremal bounds and the classification.

Check	Count
Connected base graphs, orders 2–7; optimal construction checked	995
Source partitions with 2 through $n - 1$ blocks, orders 2–7	770,021
Faithful nontrivial partitions found	201
Vertex-bound equality cases among them	37
Minimum covers used for classification checks, orders 2–5	71
Matching sign assignments checked by label-preserving isomorphism	543
Counterexamples to the stated bounds or classification	0

quotient labels. It derives the quotient directly from the source and tests the one-step transition probabilities; it does not restrict candidates to the proposed construction. For the path, triangle, and four-leaf star examples, complete projected trajectory laws were compared from every starting vertex through eight steps. Five nested expansions from a single edge produced edge counts 1, 2, 4, 8, 16, 32, and the composite projections were also checked. These finite checks supplement the proofs; they do not replace them or constitute a formal proof-assistant verification. Table 1 records the resulting counts.

Reproducibility. The verification script and its JSON output accompany the manuscript. Executing `python3 verify_random_walk_refinements.py` regenerates all reported counts using NetworkX. The program uses integer comparisons for one-step transition identities and rational arithmetic for trajectory probabilities; no numerical tolerances enter the tests.

References

- [1] M. Baker and S. Norine, *Harmonic morphisms and hyperelliptic graphs*, International Mathematics Research Notices 2009, no. 15, 2914–2955. [arXiv:0707.1309](#). The lemma numbers cited above refer to the arXiv text.
- [2] B. C. Geiger and C. Temmel, *Lumpings of Markov chains, entropy rate preservation, and higher-order lumpability*, [arXiv:1212.4375](#).